

COMSATS UNIVERSITY ISLAMABAD, LAHORE CAMPUS



OBJECT-ORIENTED PROGRAMMING

LAB TASK NO.01

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SP26-BAI-019

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BS ARTIFICIAL INTELLIGENCE (BAI-A-SP26)

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TASK NO.01: Build a class and prove independent object state

Student.java

```
class Student {  
  
    String studentId;  
  
    String name;  
  
    int completedCredits;  
  
    // Changes the state of the object it is called on  
    void addCredits(int credits) {  
        completedCredits = completedCredits + credits;  
    }  
  
    // 'degreeCredits' is a parameter, 'remaining' is a local variable  
    int remainingCredits(int degreeCredits) {  
        int remaining = degreeCredits - completedCredits;  
        return remaining;  
    }  
  
    // Returns one readable line  
    String summary() {  
        return studentId + " | " + name + " | " + completedCredits + " credits";  
    }  
}
```

Demo.java

```
public class Demo {  
  
    public static void main(String[] args) {  
        Student s1 = new Student();  
        Student s2 = new Student();  
        Student s3 = new Student();  
    }  
}
```

```
s1.studentId = "CIIT/SP26-BAI-019/LHR";
```

```
s1.name = "Faryal Sarfraz";
```

```
s1.completedCredits = 12;
```

```
s2.studentId = "CIIT/SP26-BAI-024/LHR";
```

```
s2.name = "Hania Tanveer";
```

```
s2.completedCredits = 15;
```

```
s3.studentId = "CIIT/SP26-BAI-045/LHR";
```

```
s3.name = "Sharmeen Imtiaz";
```

```
s3.completedCredits = 9;
```

```
System.out.println("--- Before change ---");
```

```
System.out.println(s1.studentId + " " + s1.name + " " + s1.completedCredits);
```

```
System.out.println(s2.studentId + " " + s2.name + " " + s2.completedCredits);
```

```
System.out.println(s3.studentId + " " + s3.name + " " + s3.completedCredits);
```

```
s2.completedCredits = s2.completedCredits + 3;
```

```
// s1 and s3 did not change because each object has its own copy of the fields.
```

```
// s2.completedCredits only modified the object that s2 refers to.
```

```
System.out.println("--- After change (only s2) ---");
```

```
System.out.println(s1.studentId + " " + s1.name + " " + s1.completedCredits);
```

```
System.out.println(s2.studentId + " " + s2.name + " " + s2.completedCredits);
```

```
System.out.println(s3.studentId + " " + s3.name + " " + s3.completedCredits);
```

```
}
```

```
}
```

Output

```
C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object Oriented Programming>javac Demo.java
C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object Oriented Programming>java Demo.java
--- Before change ---
CIIT/SP26-BAI-019/LHR Faryal Sarfraz 12
CIIT/SP26-BAI-024/LHR Hania Tanveer 15
CIIT/SP26-BAI-045/LHR Sharmeen Imtiaz 9
--- After change (only s2) ---
CIIT/SP26-BAI-019/LHR Faryal Sarfraz 12
CIIT/SP26-BAI-024/LHR Hania Tanveer 18
CIIT/SP26-BAI-045/LHR Sharmeen Imtiaz 9
```

TASK NO.02: Add behaviour

Demo2.java

```
public class Demo2 {

    public static void main(String[] args) {

        Student first = new Student();

        first.studentId = "CIIT/SP26-BAI-019/LHR";

        first.name = "Faryal Sarfraz";

        first.completedCredits = 12;


        Student second = new Student();

        second.studentId = "CIIT/SP26-BAI-024/LHR";

        second.name = "Hania Tanveer";

        second.completedCredits = 15;


        System.out.println(first.summary());

        System.out.println(second.summary());


        first.addCredits(3);

        second.addCredits(6);


        System.out.println(first.summary());
```

```

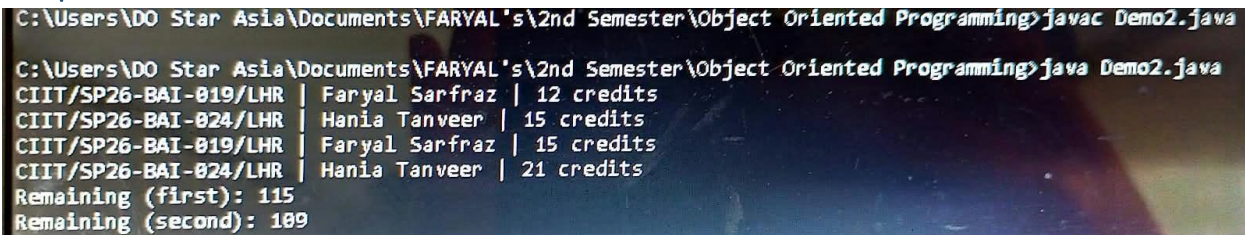
        System.out.println(second.summary());

        System.out.println("Remaining (first): " + first.remainingCredits(130));

        System.out.println("Remaining (second): " + second.remainingCredits(130));
    }
}

```

Output



```

C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object Oriented Programming>javac Demo2.java
C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object Oriented Programming>java Demo2.java
CIIT/SP26-BAI-019/LHR | Faryal Sarfraz | 12 credits
CIIT/SP26-BAI-024/LHR | Hania Tanveer | 15 credits
CIIT/SP26-BAI-019/LHR | Faryal Sarfraz | 15 credits
CIIT/SP26-BAI-024/LHR | Hania Tanveer | 21 credits
Remaining (first): 115
Remaining (second): 189

```

TASK NO.03: Method Signatures and Overloading

OverLoad.java

```

public class OverLoad {

    // Signature: enroll(String)
    void enroll(String courseCode) {

        System.out.println("Enrolled in " + courseCode);

    }

    // Signature: enroll(String, int)
    void enroll(String courseCode, int section) {

        System.out.println("Enrolled in " + courseCode + ", section " + section);

    }

    // Signature: enroll(int)
    void enroll(int numericCourseCode) {

        System.out.println("Enrolled in numeric course " + numericCourseCode);

    }

    // INVALID overload (differs only by return type).
}

```

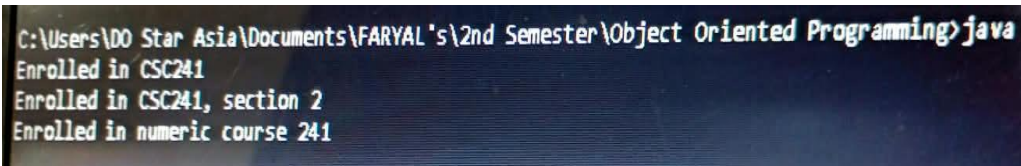
```
// Uncomment, compile, save the error to compiler_notes.txt, then comment it out again.
// int enroll(String courseCode) {
//     return 1;
// }

public static void main(String[] args) {
    OverLoad demo = new OverLoad();

    // Valid calls
    demo.enroll("CSC241");
    demo.enroll("CSC241", 2);
    demo.enroll(241);

    // Invalid calls: uncomment ONE at a time, compile, note the error, comment out again
    // demo.enroll();
    // demo.enroll("241", "2");
}
}
```

Output



```
C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object Oriented Programming>java
Enrolled in CSC241
Enrolled in CSC241, section 2
Enrolled in numeric course 241
```

TASK NO.04: Pass-By-Value

PassByValue.java

```
public class PassByValue {

    // Experiment A: Primitive

    void changeNumber(int x) {

        System.out.println("Inside method (before): x = " + x);

        x = 99;

        System.out.println("Inside method (after): x = " + x);
    }
}
```

```

}

// Experiment B: Mutate the object through the copied reference
void changeStudent(Student st) {
    System.out.println("Inside method (before): completedCredits = " + st.completedCredits);
    st.completedCredits = 99;
    System.out.println("Inside method (after): completedCredits = " + st.completedCredits);
}

// Experiment C: Reassign the parameter
void replaceStudent(Student st) {
    System.out.println("Inside method (before): name = " + st.name);
    st = new Student();
    st.name = "Temporary";
    System.out.println("Inside method (after): name = " + st.name);
}

public static void main(String[] args) {
    PassByValue demo = new PassByValue();

    // A
    int number = 10;
    System.out.println("=== Experiment A ===");
    System.out.println("Before: number = " + number);
    demo.changeNumber(number);
    System.out.println("After: number = " + number);

    // B
    Student student = new Student();
    student.name = "Faryal Sarfraz";
    student.completedCredits = 12;
    System.out.println("=== Experiment B ===");
    System.out.println("Before: completedCredits = " + student.completedCredits);
    demo.changeStudent(student);
}

```

```

        System.out.println("After: completedCredits = " + student.completedCredits);

        // C

        System.out.println("=== Experiment C ===");

        System.out.println("Before: name = " + student.name);

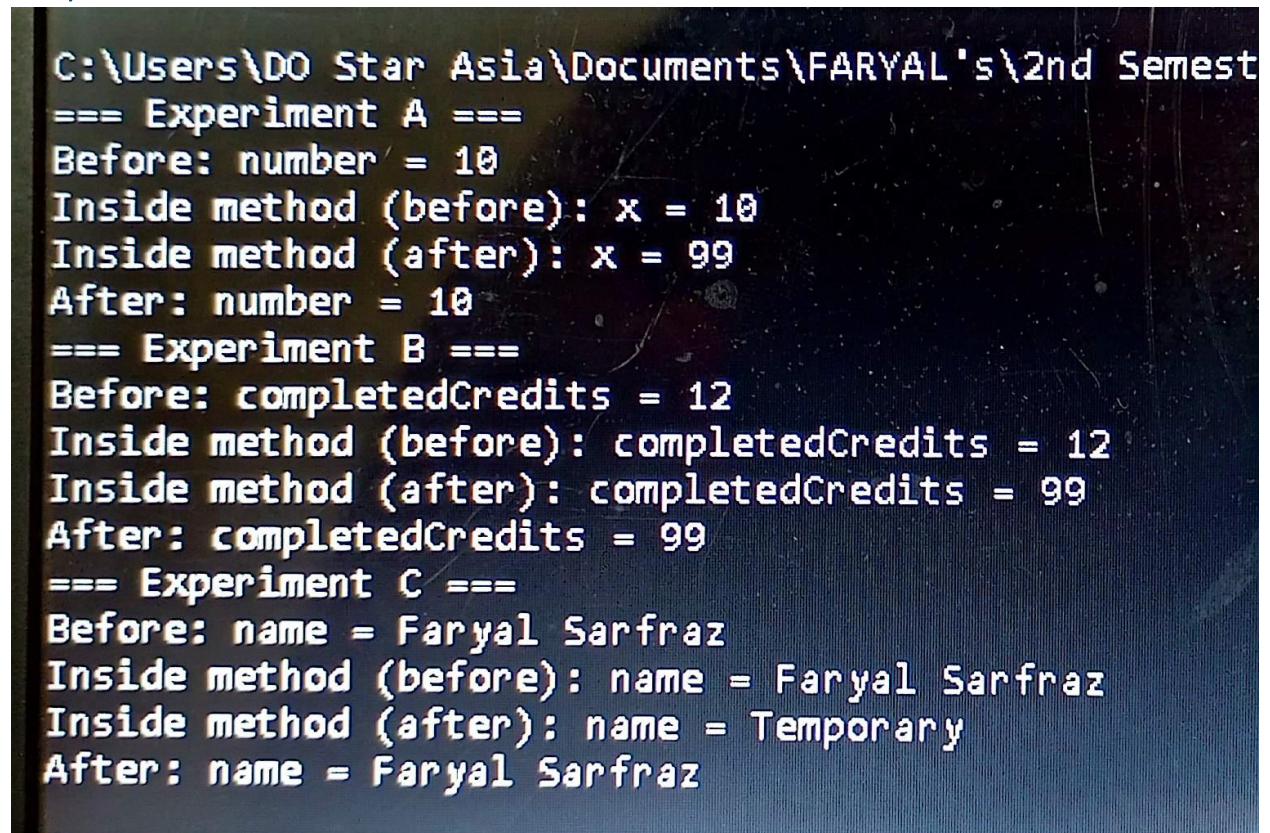
        demo.replaceStudent(student);

        System.out.println("After: name = " + student.name);

    }
}

```

Output



```

C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semest
=== Experiment A ===
Before: number = 10
Inside method (before): x = 10
Inside method (after): x = 99
After: number = 10
=== Experiment B ===
Before: completedCredits = 12
Inside method (before): completedCredits = 12
Inside method (after): completedCredits = 99
After: completedCredits = 99
=== Experiment C ===
Before: name = Faryal Sarfraz
Inside method (before): name = Faryal Sarfraz
Inside method (after): name = Temporary
After: name = Faryal Sarfraz

```

TASK NO.05: AI Experiment

AI.java

```
class AI {
```

```
    String experimentName;
```

```

int completedEpochs;

int targetEpochs;

// Changes state
void runEpochs(int epochs) {
    completedEpochs = completedEpochs + epochs;
}

// Overload: signature runEpochs(int, int)
void runEpochs(int epochs, int bonusEpochs) {
    int totalEpochs = epochs + bonusEpochs; // local variable
    completedEpochs = completedEpochs + totalEpochs;
}

// Only computes and returns information
int remainingEpochs() {
    return targetEpochs - completedEpochs;
}

String status() {
    return experimentName + ": " + completedEpochs + "/" + targetEpochs
        + " epochs done, " + remainingEpochs() + " remaining";
}
}

```

AlDemo.java

```

public class AlDemo {

    // Helper method: receives a copy of the reference value
    void addHelperEpochs(Al experiment) {
        experiment.completedEpochs = experiment.completedEpochs + 5;
    }

    public static void main(String[] args) {
        AlDemo demo = new AlDemo();
    }
}

```

```
AI cnn = new AI();  
cnn.experimentName = "CNN Image Classifier";  
cnn.completedEpochs = 0;  
cnn.targetEpochs = 50;
```

```
AI lstm = new AI();  
lstm.experimentName = "LSTM Text Model";  
lstm.completedEpochs = 10;  
lstm.targetEpochs = 30;
```

```
System.out.println(" Initial state ");  
System.out.println(cnn.status());  
System.out.println(lstm.status());
```

```
cnn.runEpochs(10);  
cnn.runEpochs(5, 2);
```

```
System.out.println(" After running only cnn ");  
System.out.println(cnn.status());  
System.out.println(lstm.status());  
System.out.println("cnn remaining: " + cnn.remainingEpochs());  
System.out.println("lstm remaining: " + lstm.remainingEpochs());
```

```
System.out.println(" Pass-by-value mutation ");  
System.out.println("Before helper: " + lstm.completedEpochs);  
demo.addHelperEpochs(lstm);  
System.out.println("After helper: " + lstm.completedEpochs);
```

```
}  
}
```

Output

```
C:\Users\DO Star Asia\Documents\FARYAL's\2nd Semester\Object  
Initial state  
CNN Image Classifier: 0/50 epochs done, 50 remaining  
LSTM Text Model: 10/30 epochs done, 20 remaining  
After running only cnn  
CNN Image Classifier: 17/50 epochs done, 33 remaining  
LSTM Text Model: 10/30 epochs done, 20 remaining  
cnn remaining: 33  
lstm remaining: 20  
Pass-by-value mutation  
Before helper: 10  
After helper: 15
```